

Level 0a Solving linear equations

(How to think about system $Ax=b$ and its solution set)

players · vectors in $\mathbb{R}^n$, written as columns (sometimes rows)

· Matrices $A_{r \times c}$ thought as defining a function from $\mathbb{R} \rightarrow \mathbb{R}$

questions · How big is the solution space?

$$x \mapsto Ax$$

· How to find, how to describe conceptually? Any structure?

Answers · row reduction, pivots

· general solution as a translate

Behavior of function $f(x) = Ax$

What does injectivity/surjectivity

mean for — solutions of relevant system?

— rows of A ?

— columns of A ?

— RREF of A ?

Slogans

· Think geometrically!

· $Ax = \left\{ \begin{array}{l} \text{linear combination of} \\ \text{columns of } A \text{ with wts from } x \end{array} \right.$

· Row operations are

— reversible, do not change solⁿ space

— equivalent to left multiplication by —?

— don't change row space, don't change dependencies among columns

Level 0B

Level 0B (abstract) Vector spaces

Words: linear combination, span, independence, basis

players

- linear maps between vector spaces
- parametrization by matrices after choosing bases

questions

- how different bases talk to each other (penitence for violence)
- what is the structure of a linear map?

answers

- dimension and its invariance

- formulas for change of basis: for a vector space, for a linear map

Slogans

- Geometry Continues!
- uniformity of fibers

- rank-nullity thm

- row rank = col rank

Level 1 Eigenanalysis

just basis: for a vector space, for a linear map $\cdot$ row rank = col rank

Level 1 Eigenanalysis understanding the soul of an operator

Level 2 Length and perpendicularity (over $\mathbb{R}$ and $\mathbb{C}$)

2A dot product $\rightsquigarrow$ 2B inner products

The pinnacle: the spectral theorem!